SRS • VERSION 1.0

Polymarket BTC AI Trader

Software Requirements Specification (SRS)

PRODUCT TYPE	INITIAL MARKET	TRADING MODES
AI-assisted prediction-market trading software	Polymarket BTC short-duration markets	Manual approval and automated execution

Prepared as a practical concept draft for planning, developer discussion, and staged validation.

Important: This project cannot guarantee a high win rate or profits. All trading is risky. The first release should use historical testing and paper trading before any live automated orders.

Document Control

Field	Value
Product	Polymarket BTC AI Trader
Version	1.0 — draft for developer review
Date	11 October 2026
Primary interface	Responsive web app / PWA
Initial account model	Single owner account; no customer funds custody
Primary modes	Manual approval and automated execution
Initial markets	Discoverable BTC-related Polymarket markets, including short-duration markets where available

1. Purpose

This SRS defines the functional, non-functional, security, data, testing and acceptance requirements for a software application that analyzes Polymarket BTC prediction markets and supports manual or automated order execution. It is a draft baseline for implementation; endpoint details must be confirmed against the current official Polymarket API documentation before coding.

2. Scope

In scope: dynamic market discovery; market data ingestion; model-based probability estimates; rule-based trade filtering; manual and automatic modes; risk controls; authenticated execution; portfolio and order tracking; audit logs; paper trading; historical testing; dashboard and alerts.

Out of scope for initial release: operating a prediction market; accepting or pooling deposits from users; custody of third-party funds; guaranteeing returns; bypassing geographic or account restrictions; autonomous changes to risk limits; multi-user SaaS until a separate legal/security review.

3. Definitions

Term	Definition
Market	A Polymarket event/contract with defined outcomes and resolution rules.
Outcome token	A token/share representing a market outcome, as implemented by the platform.
CLOB	Central Limit Order Book interface used for market data and order operations.
Manual mode	A proposed trade requires explicit user approval before submission.
Auto mode	Qualifying trades may be submitted automatically within preconfigured limits.
Paper trade	Simulated order/trade that does not submit a real order or risk funds.
Expected value	Modelled expected return using probability and executable price, adjusted for costs and uncertainty.
Kill switch	Control that prevents new orders and, according to configuration, cancels open orders.

4. User Roles

- Owner: configures strategies, risk limits, mode, account connection and emergency stop.
- Viewer (optional later): can view dashboards without placing orders or changing settings.
- System service: ingests data, evaluates strategy, executes approved orders, reconciles status and records logs.

5. Functional Requirements

FR-01 — Market discovery

The system shall discover current eligible BTC-related markets from the supported official Polymarket market-data interfaces. It shall not hardcode a single five-minute market ID or assume a market exists continuously.

FR-02 — Contract metadata

For each candidate market, the system shall store market identifier, question, outcomes, outcome/token identifiers, open/closed status, start/end or expiry timestamps where provided, strike/reference, resolution criteria, and source URL when available. Ambiguous or missing critical metadata shall block live trading.

FR-03 — Market data

The system shall ingest relevant Polymarket prices, order book, spread, liquidity and trades where supported, plus an independent BTC price feed. Each observation shall include source and timestamp. Data exceeding a configurable freshness limit shall be marked stale and shall not trigger live orders.

FR-04 — Probability engine

The system shall generate a probability estimate and uncertainty indicator for each candidate outcome. Model version, input snapshot, decision timestamp and output shall be recorded for audit and evaluation.

FR-05 — Opportunity filter

The system shall estimate net expected value using an executable price and configurable assumptions for fees, spread, slippage and partial fills. It shall reject trades that fail the configured minimum edge, data-quality, liquidity or risk conditions.

FR-06 — Manual mode

The dashboard shall display a proposed trade with market, side/outcome, limit price, stake, estimated cost, remaining time, probability estimate, expected value, major risks and expiry. The system shall require a deliberate approval action before submitting the order. Approval shall expire if the market or price changes beyond configured thresholds.

FR-07 — Auto mode

Auto mode shall be disabled by default. When enabled by the owner, the system may submit qualifying orders only if all risk and data checks pass. Switching to Manual shall stop new autonomous submissions; open orders shall be handled according to a clearly displayed cancellation/management policy.

FR-08 — Order execution

The execution service shall support the officially documented order types and operations available to the account. It shall validate order parameters, use idempotency/client identifiers where supported, avoid duplicate submissions on retry, track accepted/rejected/partially filled/filled/cancelled states, and preserve exchange/API response references.

FR-09 — Position and balance tracking

The system shall display account collateral/balance and position information only from authorized supported sources. It shall distinguish available balance, reserved/open-order funds, positions, realized P&L;, unrealized P&L; and settlement state where the API makes these available.

FR-10 — Reconciliation

At startup and periodically, the system shall reconcile local order/position records with platform state. If a material mismatch is detected, it shall pause new live orders, log the discrepancy and notify the owner.

FR-11 — Risk limits

- Owner-configurable maximum stake per trade and maximum open exposure.

- Daily loss limit and drawdown circuit breaker.
- Maximum number of orders/trades per time window.
- Market liquidity and spread thresholds.
- No trading when account status, location eligibility, market resolution criteria, price feed or API state cannot be verified.
- No increase in risk limits by the model itself.
- Kill switch to block new orders and optionally cancel open orders; the UI must state what the switch does.

FR-12 — Paper trading and backtesting

The system shall support simulated execution without live orders. Backtesting shall use time-ordered data, avoid future information, include configurable transaction costs and slippage, and report results on separate training, validation and test periods.

FR-13 — Performance reporting

The system shall report number of trades, win rate, average win/loss, net P&L;, fees, profit factor, expectancy, maximum drawdown, calibration metrics, rejected signals and performance by market/time-to-expiry. It shall clearly label simulated versus live results.

FR-14 — Audit trail

The system shall store timestamped market snapshots or reproducible references, model version, signal, decision reason, risk checks, user approvals, order request/result, fills, fees, cancellations, reconciliations and configuration changes. Secrets must never be written to logs.

FR-15 — Alerts

The system shall alert the owner to order failures, stale data, risk-limit trips, API authentication errors, reconciliation discrepancies, unexpected market closure and system outages. Notifications shall not expose credentials or unnecessary sensitive data.

FR-16 — Deposits and withdrawals

Version 1 shall not process or custody deposits/withdrawals. The user shall deposit and withdraw through Polymarket's official account interface or supported route. The app may show available account information and provide navigation to official account pages, but must not claim that a transfer completed without authoritative confirmation.

6. Non-Functional Requirements

Area	Requirement
Security	TLS in transit; encrypted secret storage; backend-only API credentials; least privilege; redacted logs; key rotation; no secrets in client bundles.
Availability	Graceful recovery after process/network restart; reconnect with bounded exponential backoff; pause live trading on uncertain state.
Performance	Measure end-to-end data-to-decision and order-submission latency; do not promise ultra-low latency without deployment measurements.
Reliability	Use durable order/event records, idempotent retries where possible, and reconciliation before resuming after outages.
Usability	Responsive dashboard; explicit Manual/Auto status; confirmation for dangerous actions; clear display of simulation versus live mode.
Auditability	Every live decision and order must be traceable to inputs, model version, risk checks and user/system action.

Area	Requirement
Maintainability	Separate market-data, strategy, risk, execution, portfolio, UI and notification modules.
Privacy	Collect minimal user data; define retention and deletion policies; restrict access to account activity.
Compliance	Verify current platform terms, API policies, geographic eligibility and applicable legal obligations before live trading or SaaS release.

7. High-Level Architecture

Component	Suggested responsibility
Web/PWA Frontend	Dashboard, charts, mode toggle, trade approval, settings, alerts and reports.
Backend API	Authentication, authorization, dashboard data, configuration validation and request coordination.
Market Data Worker	Polymarket market discovery/CLOB data and independent BTC price feed; timestamp validation.
Strategy Service	Feature calculation, baseline rules, model inference and candidate trade generation.
Risk Engine	Independent final gate for exposure, liquidity, stale data, loss limits and eligibility flags.
Execution Service	Polymarket authenticated order placement/cancellation and order-state updates.
Portfolio/Reconciliation Worker	Balance/position synchronization, fill tracking, settlement updates and discrepancy handling.
Database	Relational records for users/configuration, market metadata, signals, orders, fills, positions, risk events and audit logs.
Secret Store	Encrypted credentials and controlled access by the execution service only.

8. Suggested Data Model

Entity	Important fields
users	id, email, created_at, status
exchange_connections	id, user_id, provider, encrypted_secret_reference, status, last_verified_at; never store secrets in plaintext
strategy_configs	id, user_id, mode, model_version, min_edge, max_stake, max_exposure, daily_loss_limit, enabled
markets	id, provider_market_id, question, status, outcomes_json, token_ids_json, start_at, end_at, resolution_rules, updated_at
market_snapshots	id, market_id, observed_at, source, prices_json, orderbook_json, liquidity, spread, freshness_status
signals	id, market_id, created_at, model_version, probability_json, candidate_side, executable_price, expected_value, rationale_json, rejected_reason
trade_approvals	id, signal_id, user_id, approved_at, expires_at, approved_parameters_json
orders	id, provider_order_id, market_id, signal_id, mode, side, limit_price, quantity, status, submitted_at, updated_at, idempotency_key
fills	id, order_id, provider_fill_id, price, quantity, fee, filled_at
positions	id, market_id, outcome, quantity, average_cost, status, last_reconciled_at
risk_events	id, user_id, event_type, severity, occurred_at, details_redacted_json
audit_logs	id, actor_type, actor_id, action, occurred_at, correlation_id, details_redacted_json
performance_snapshots	id, period_start, period_end, mode, trade_count, net_pnl, drawdown, win_rate, metrics_json

9. Key Workflows

9.1 Candidate-to-order workflow

1. Discover market and validate status, outcomes, expiry and resolution criteria.
2. Collect fresh Polymarket data and independent BTC data; reject incomplete/stale inputs.
3. Calculate features and estimate outcome probabilities.
4. Calculate expected value at an executable price after costs and uncertainty margin.
5. Apply risk engine; if any hard gate fails, log the reason and do not trade.
6. Manual mode: present approval request and expire it if conditions change. Auto mode: submit only when configured conditions still hold.
7. Track order status and fills, reconcile platform state, and record the full decision trail.

9.2 Restart or API outage workflow

1. Set live execution to paused when state is uncertain.
2. Reconnect with bounded retries and verify authentication.
3. Fetch current open orders, positions and balances from supported sources.
4. Reconcile with local records and handle duplicates/missing events.
5. Resume only after checks pass; otherwise notify owner and remain paused.

10. Security Requirements

- Use a supported secure authentication flow; keep signing secrets on the server only.
- Do not request a wallet seed phrase through the website or chat.
- Use separate development and production credentials where supported.
- Protect dashboard actions with authentication and CSRF/session safeguards appropriate to the stack.
- Rate-limit sensitive endpoints; validate all order inputs server-side.
- Require explicit owner action to enable live Auto mode and change risk limits.
- Provide a visible emergency stop and log its use.
- Test for secret leakage, unauthorized order placement, replay/duplicate orders and stale-price execution.

11. Acceptance Criteria

ID	Pass condition
AC-01	Market discovery identifies current supported BTC markets dynamically and does not rely on a hardcoded five-minute ID.
AC-02	Markets with missing/ambiguous resolution criteria or stale inputs are blocked from live trading.
AC-03	Manual mode cannot submit an order before a valid user approval; stale approvals are rejected.
AC-04	Auto mode remains disabled by default and cannot bypass any risk-engine hard gate.
AC-05	Duplicate retries do not create duplicate orders where supported by API/client idempotency design.
AC-06	Order status, fills, fees, positions and balance information reconcile with supported platform data.
AC-07	Daily loss/exposure limits trigger a pause and notify the owner.
AC-08	Kill switch blocks new submissions and follows the documented open-order policy.

ID	Pass condition
AC-09	Backtests and paper trades are visibly separated from live results and include realistic costs.
AC-10	No API secrets appear in frontend assets, logs, URLs or error messages.
AC-11	The user can export a trade history and inspect why a signal was accepted or rejected.
AC-12	Deposit and withdrawal features do not imply custody; the user is directed to official Polymarket flows.

12. Testing Plan

- Unit tests: market parsing, expiry handling, feature calculations, probability math, fee assumptions, sizing and risk gates.
- Integration tests: API connectivity, authentication failure, order rejection, partial fill, cancellation and data reconnect.
- Replay/backtest tests: different volatility regimes, market gaps, thin liquidity, fast price moves and end-of-window behavior.
- Paper-trading tests: live data with simulated fills; compare assumed and observed spreads and latency.
- Security tests: secret exposure, permission checks, duplicate order requests, session protection and unauthorized configuration changes.
- Failure injection: disconnect data feeds, restart workers, return API errors, introduce stale timestamps and simulate mismatched positions.

13. Developer Deliverables

- Working source code and reproducible setup instructions.
- Architecture and data-flow diagrams or equivalent technical documentation.
- Environment variable template with placeholders only; no real credentials in source control.
- Database migrations and seed/demo data.
- Unit/integration test suite and sample test report.
- Backtest and paper-trading reports that clearly disclose assumptions.
- Deployment guide, backup/recovery plan, logging and monitoring instructions.
- User guide explaining modes, risk limits, deposits/withdrawals through Polymarket, and emergency stop.

14. Open Decisions Before Development

- Confirm the exact currently available BTC market types, durations and resolution rules from live official market data.
- Confirm Polymarket API authentication, order types, collateral/token handling, fees and rate limits from current docs.
- Confirm account and geographic eligibility for the owner and target users.
- Select baseline strategy and model after data audit; do not pre-commit to a claimed win rate.
- Choose backend language, database, hosting and alert provider.
- Decide whether the first MVP is personal-use only; treat public SaaS as a separate phase.

15. Official Documentation

- https://docs.polymarket.com/
- https://docs.polymarket.com/developers/CLOB/introduction

- https://data-api.polymarket.com/
- Check current Polymarket terms, geographic restrictions, fee schedule and API documentation before live deployment.

End of SRS. This specification is a development baseline and does not guarantee trading performance or replace current platform/legal verification.